Joint Classification of Hyperspectral and LiDAR Data Using Height Information Guided Hierarchical Fusion-and-Separation Network

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***Abstract*— Hyperspectral image (HSI) and light detection and ranging (LiDAR) data are complementary to each other, which can be combined to improve the classification performance. However, existing deep network models do not sufficiently consider their complementarity to design the network structure and loss functions. Moreover, there lacks a hierarchical mutual-assistance learning mechanism that leverages the modality-shared features to enhance the modality-specific ones and vice versa. In view of these, we propose a novel height information guided hierarchical fusion-and-separation network (HFSNet) for joint classification of HSI and LiDAR data. HFSNet consists of three major components, i.e., dual-structure feature encoders (DSFEs), feature fusion-and-separation blocks (F2SBs), and an edge decoder (ED). Specifically, the transformer and convolutional neural network (CNN) are introduced in DSFEs to encode the spectral and spatial information of HSI and LiDAR data, respectively. In F2SBs, the deformable convolution-based height information guided fusion module (HIGFM) and the modality separation refinement module (MSRM) are proposed to sequentially extract modality-shared and modality-specific features. Additionally, the ED is incorporated into our model to predict the LiDAR edge map from the HSI feature to improve the model's generalization ability. As such, the learned features from HSI and LiDAR data are deeply fused and mutually enhanced. Experiments on three benchmark datasets show the superiority of HFSNet to the state-of-the-art methods for jointly classifying HSI and LiDAR data with limited training samples.**

***Index Terms*— Classification, deep learning, hyperspectral image (HSI), light detection and ranging (LiDAR) data, transformer.**

NOMENCLATURE

$X_h^0 \in \mathbb{R}^{C\times N\times N}$	Input HSI cube.
C	Number of spectral dimensions.
$N \times N$	Spatial size.
$X_h^i \in \mathbb{R}^{S_i\times N\times N}$	Input of the ith transformer-based spectral encoder ($i = 1, 2, 3$).
$F_h^i \in \mathbb{R}^{P_i\times N\times N}$	Output of the ith transformer-based spectral encoder ($i = 1, 2, 3$).
$X_l^i \in \mathbb{R}^{S_i\times N\times N}$	Input of the ith CNN-based spatial encoder ($i = 1, 2, 3$).
$F_l^i \in \mathbb{R}^{P_i\times N\times N}$	Output of the ith CNN-based spatial encoder ($i = 1, 2, 3$).
$X_s^1 \in \mathbb{R}^{P_1\times N\times N}$	Input tensor with all elements being zero in the first F2SB.
$X_s^{i+1} \in \mathbb{R}^{P_i\times N\times N}$	Modality-shared feature in the ith F2SB ($i = 1, 2, 3$).
L	Number of repeated operations in each transformer-based spectral encoder.
$\hat{X}_{\text{edge}} \in \mathbb{R}^{N\times N}$	Output of ED (i.e., the predicted edge map).

I. INTRODUCTION

HYPERSPECTRAL images (HSIs), which usually consist of hundreds of consecutive narrow bands, provide rich spectral information to identify lots of land cover types. HSIs have been applied in many fields, including ocean hydrographic detection [1], precision agriculture [2], environmental monitoring [3], and so on. For HSI classification, a large number of methods have been proposed in the literature [4], [5], [6], [7], [8]. Despite the great progress, it is still challenging to improve the classification accuracy of HSIs in complex environments. For example, the ground and roof are composed of the same concrete mixes. Due to the similar spectral properties, it is difficult to distinguish these two objects

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based on spectral information alone. With the development of remote sensing technologies, the acquisition of multisource images has become relatively easy, making the extraction of complementary information possible. For example, light detection and ranging (LiDAR) data provide height information, panchromatic images provide spatial information, while thermal infrared images provide rich temperature information. The use of multimodal data to obtain complementary information has become one of the important research tops in the remote sensing field [9], [10].

In the early studies of HSI and LiDAR data classification, most methods apply machine learning to extract handcrafted or shallow features. For example, by combining HSI and LiDAR data, Dalponte et al. [11] used support vector machines and Gaussian maximum likelihood methods to classify complex forest areas. Pedergnana et al. [12] employed extended morphological attribute profiles (EAPs) to classify optical and LiDAR Data. Ghamisi et al. [13] used attribute profiles (APs) to model the spatial information of HSI and LiDAR data, and fed the fused features into the classifier to output classification results. These fusion methods which directly stack multiple features with a limited number of labeled samples can lead to the curse of dimensionality. To solve this problem, Liao et al. [14] proposed a generalized graph-based fusion method to couple dimension reduction and feature fusion of the spectral information and morphological profiles. In addition, orthogonal total variation component analysis [15] and sparse and low-rank component analysis (SLCA) [16] were proposed to reduce the dimensionality of fused features.

The above handcrafted methods are difficult to extract deep and nonlinear features, and show limited performance improvements. To solve this problem, various deep learning-based frameworks have been proposed. For example, the 2-D convolution was used in [17] and [18] to extract spatial features, and 1-D convolution was used in [19] to extract spectral features. The 3-D convolution was also explored in [20] and [21] to simultaneously extract the spatial and spectral information. Besides, Wu et al. [22] developed two convolutional neural network (CNN) models to extract information from HSI and LiDAR data, and designed a reconstruction strategy to learn more compact feature representations. Zhao et al. [23] proposed a hierarchical random walk network, which jointly optimized the dual-tunnel CNN and pixelwise affinity via a novel random walk layer. Du et al. [24] concatenated heterogeneous data at the input layer and utilized a multimodal graph to guide the feature extraction process for classification. To exploit different receptive fields, Xue et al. [25] proposed a multiscale deep learning network with self-calibrated convolution. Furthermore, Fang et al. [26] proposed a spatial–spectral enhancement network (S^2ENet) by considering cross-modal information interactions in deep neural networks. Mohla et al. [27] proposed a feature fusion and extraction framework called attention based multimodal fusion network (FusAtNet), which used the self-attention mechanism to highlight spectral features of HSIs and spatial features of LiDAR data, and further used the spatial features to weight HSI features.

Although CNN-based networks have achieved good classification results, they are inadequate to capture the global context dependencies of spectral sequences. To overcome this issue, transformers [28] have been recently introduced in the remote sensing field. For example, He et al. [29] proposed the bidirectional encoding representation from transformers for HSI classification. Hong et al. [30] constructed a transformer-based backbone network, which can be applicable to both pixelwise and patchwise inputs. Zhong et al. [31] presented a spectral-spatial transformer consisting of spatial attention and spectral association modules. Transformers tend to ignore local spatial information. To combine the advantages of CNN and transformer, Wang et al. [32] designed a multiscale convolutional transformer network to encode spatial and spectral information. Zhao et al. [33] proposed a convolutional transformer by fusing local and global features. Yang et al. [34] developed an HSI transformer by embedding a correlated convolutional module to capture subtle spectral differences and local spatial information. Song et al. [35] used the 3-D convolution and transformer to jointly extract spatial and spectral features of HSIs. Nowadays, few transformer-based works have been used for joint classification of HSI and LiDAR data. For example, Xue et al. [36] utilized three branches of transformers to extract spectral and spatial features from HSI and LiDAR data, and all the heterogeneous features were finally fused at the last level. Roy et al. [37] introduced a multimodal fusion transformer (MFT) where the input patches were taken from the HSI while the class token was generated from other multimodal data like LiDAR. Zhao et al. [38] combined the CNN and transformer structures to build a dual-branch network where a cross-token attention fusion encoder was used to fuse the final features.

The aforementioned methods focus on fusing HSI and LiDAR data at a particular network layer. However, due to the heterogeneous characteristics of multimodal data and hierarchical properties of feature representations in different network layers, it is necessary to fuse multilevel features with deep interactions. To this end, Li et al. [39] proposed a dual-channel spatial, spectral and multiscale attention convolutional long short-term memory (ConvLSTM) neural network with a three-level fusion strategy. Dong et al. [40] proposed a multibranch feature fusion network by exploiting the hierarchical features extracted from HSI and LiDAR level by level. Wang et al. [41] presented a multiattentive hierarchical dense fusion network (MAHiDFNet) to realize the feature-level fusion and classification of HSI and LiDAR data. Lu et al. [42] proposed a fusion classification method including a coupled adversarial feature learning subnetwork and a multilevel feature fusion classification subnetwork. Wang et al. [43] introduced a spectral-spatial mutual-guided module for the collaborative transmission of spectral features and spatial information. Despite these developments, the above approaches do not well consider the refinement of modality-specific features using the fused modality-shared ones, which may limit the model's representation ability. Also, the advanced transformer is not considered for spectral sequence modeling. Hence, it remains a challenge to build a deep cross-modality information

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interaction network which not only fully considers heterogeneous characteristics of multimodal data but also well handles modality-shared and modality-specific features to boost each other.

Motivated by the above works, in this article, we propose a height information guided hierarchical fusion-and-separation network (HFSNet) for joint classification of HSI and LiDAR data. In the proposed HFSNet, we fully explore the heterogeneous properties of HSI and LiDAR data to design our network structure and loss functions. We introduce a hierarchical mutual-assistance learning mechanism that leverages the modality-shared features to enhance the modality-specific ones and vice versa. Experiments show the superiority of HFSNet to the state-of-the-art methods for joint classification of HSI and LiDAR data with limited training samples. The main contributions of this work are summarized as follows.

1) We design dual-structure feature encoders (DSFEs), including transformer-based spectral encoders and CNN-based spatial encoders, to extract complementary features from HSIs and LiDAR data. Specifically, by taking into account the input data's characteristics, we effectively encode long-range spectral sequence information of HSIs and local spatial information of LiDAR images with the help of DSFEs. We also embed the Laplacian filter in spatial encoders to excavate the useful height (elevation) information contained in the LiDAR images.
2) We introduce hierarchical feature fusion-and-separation blocks (F2SBs), each containing a height information guided fusion module (HIGFM) and a modality separation refinement module (MSRM), to promote the mutual learning of modality-shared and modality-specific features. Specifically, HIGFM leverages the height information of LiDAR data via deformable convolutions to guide the fusion of dual-modality data, while MSRM separates modality-specific features from the fused ones. As a result, these two modules achieve cooperative learning and mutual promotion in a deeply interactive way.
3) We build an edge decoder (ED) on the last output HSI feature to predict the binary LiDAR edge map. This module acts as an auxiliary learning task, which performs task-level cross-modality information interactions to improve the model's generalization ability. In addition, ED is only used for training, and hence the classification speed will not be affected at inference time.

Our work is related to [44], [45], and [46], but there are several major differences. First, elementwise subtraction operations are used in these three methods to compute differential features for cross-modality information interactions, but the modality-shared features are not generated [44] or well explored [46]. In comparison, our method leverages deformable convolutions to elegantly fuse heterogeneous data and obtain modality-shared features. Second, unlike [45] that focuses only on the fusion process, our method further uses the fused features to separate and refine modality-specific ones, and hence develops a mutual-assistance learning mechanism to promote the model's representation ability. Third, our method differs from [44], [45], and [46] in network structures, loss functions and visual tasks. In particular, our method employs the complementary CNN and transformer as the network structure, and designs an auxiliary loss function to improve the model's generalization ability.

The rest of this article is organized as follows. The proposed network is introduced in Section II. The classification experiments are presented in Section III. Finally, the conclusions are drawn in Section IV.

II. Architecture of HFSNet for Joint Classification of HSI and LiDAR Data

Fig. 1 shows the flowchart of HFSNet for joint classification of HSI and LiDAR data, which mainly consists of three components: DSFEs, F2SBs, and an ED. The HSI stream and the LiDAR stream represent modality-specific information stream. They interact with each other in the shared stream via the hierarchal F2SBs. The ED is only used in the training phase for auxiliary learning. The notations used in this article are listed in Nomenclature. Each component is detailed in Sections II-A–II-D

A. Dual-Structure Feature Encoders

The HSI and LiDAR data (in this article both are images) have distinct characteristics. That is, the HSI has an abundance of spectral sequence information, while the LiDAR image has only one channel and the spatial size of input patches is small. Hence, we use the transformer which is good at modeling long-range dependencies to capture the relationships between spectral sequence data, and use the CNN to capture the spatial information of LiDAR images. Fig. 1 shows the multilevel structure of DSFEs, each containing a transformer-based spectral encoder and a CNN-based spatial encoder.

1) Transformer-Based Spectral Encoder: As HSIs contain redundant spectral sequence information, we first use group-wise convolutions to reduce the number of spectral bands. Let $X_h^0 \in \mathbb{R}^{C\times N\times N}$ be the input HSI cube, where $N \times N$ is the spatial size and C is the number of spectral dimensions. We pass X_h^0 to three parallel group convolutions (GCs), which have different numbers of groups in each GC, and add up the results to obtain the feature $X_h^1 \in \mathbb{R}^{S_1\times N\times N}$, i.e.,

$$X_h^1 = \mathrm{GC}\left(X_h^0, n_1\right) + \mathrm{GC}\left(X_h^0, n_2\right) + \mathrm{GC}\left(X_h^0, n_3\right) \quad (1)$$

where n_1, n_2 and n_3 are the number of groups in the three GCs (we set $n_1 = 1$, $n_2 = 2$, $n_3 = 4$). Without loss of generality, we denote the input of the ith transformer-based spectral encoder as $X_h^i \in \mathbb{R}^{S_i\times N\times N}$ $(i = 1, 2, 3)$, as shown in Fig. 1.

Then, we reshape X_h^i into a set of HSI tokens $X = [x_1, \ldots, x_{S_i}] \in \mathbb{R}^{n\times S_i}$, where $n = N \times N$ is the embedding dimension and S_i is the number of tokens. After the position embedding (PE) and LayerNorm (LN), we send the result into the multihead self-attention (MHSA) to capture global spectral sequence information, i.e.,

$$X' = \mathrm{PE}(X) + X \quad (2)$$

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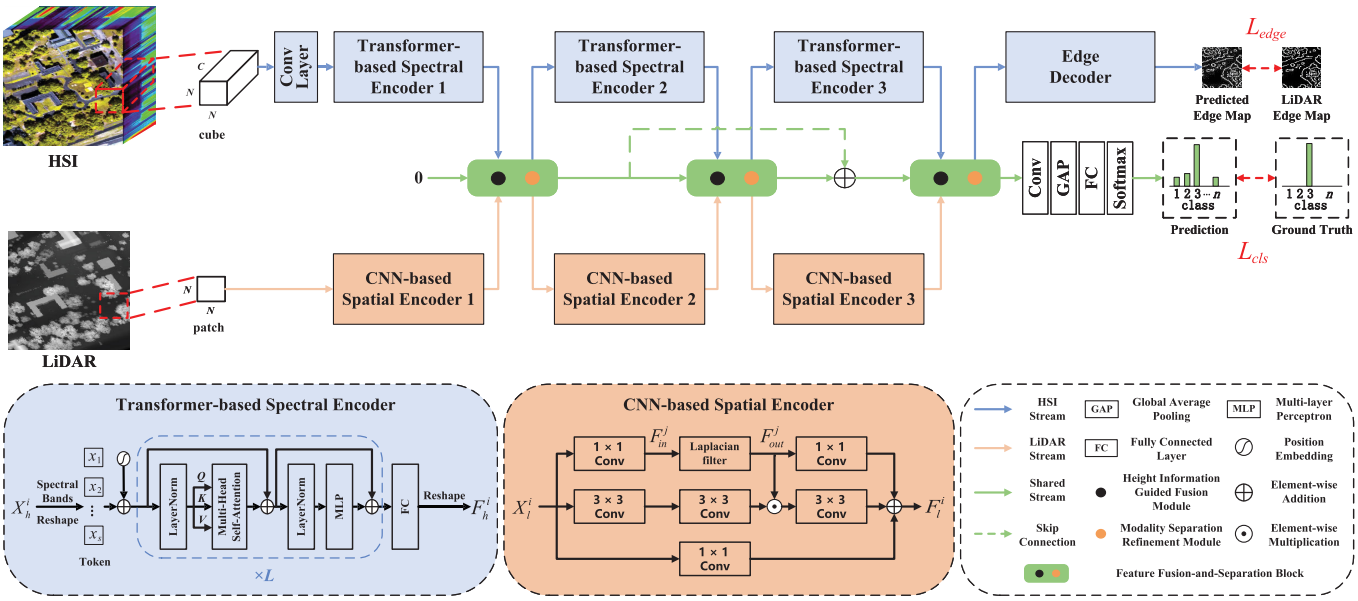


Fig. 1. Flowchart of the proposed HFSNet for joint classification of HSI and LiDAR data (the ED is only used in the training phase).

$$Z = \mathrm{LN}(X') \tag{3}$$

$$\mathrm{MHSA}(Z) = \mathrm{Concat}(\mathrm{SA}_1, \mathrm{SA}_2, \ldots, \mathrm{SA}_h)W \tag{4}$$

$$\mathrm{SA}_j = \mathrm{Softmax}\left(\frac{Q_j K_j^{\mathsf{T}}}{\sqrt{d}}\right)V_j \tag{5}$$

where "Concat" denotes the concatenation operation, h is the number of attention heads, $Q_j = ZW_j^Q$, $K_j = ZW_j^K$, $V_j = ZW_j^V$ are query, key and value, respectively ($j = 1, \ldots, h$), W_j^Q, W_j^K, $W_j^V \in \mathbb{R}^{n\times d}$ are three learnable weight matrices ($d = n/h$), and $W \in \mathbb{R}^{h\times d\times S_i}$ is a transformation matrix for aligning the feature dimension.

Next, we follow MHSA with an LN and a multilayer perception (MLP) to obtain the following outputs:

$$Z' = \mathrm{MHSA}(Z) + X' \tag{6}$$

$$Z'' = \mathrm{MLP}(\mathrm{LN}(Z')) + Z'. \tag{7}$$

As shown in Fig. 1, the above procedure is repeated L times. Here, we set L to 2, 2, 6 for the first, the second and the third transformer-based spectral encoder, respectively.

Finally, we use linear projection implemented with a fully connected layer to adjust the number of output channels, and reshape Z'' to $F_h^i \in \mathbb{R}^{P_i\times N\times N}$ as the output of the ith transformer-based spectral encoder.

2) CNN-Based Spatial Encoder: For LiDAR images, we adopt CNN-based spatial encoders to extract hierarchal features. Particularly, we embed the Laplacian filter to explicitly extract edge features and encode the height (elevation) information. Fig. 1 shows the detailed structure of the CNN-based spatial encoder.

In Fig. 1, $X_l^i \in \mathbb{R}^{S_i\times N\times N}$ denotes the input of the ith CNN-based spatial encoder ($i = 1, 2, 3$), where $N \times N$ is the spatial size of LiDAR patch/feature, and S_i is the number of channels (especially $S_1 = 1$). In each encoder, we employ multiple 1×1 convolutional layers to adjust the number of channels, and use a Laplacian filter to extract edge features

$$W_{\mathrm{lap}} = \begin{bmatrix} +1 & +1 & +1 \\ +1 & -8 & +1 \\ +1 & +1 & +1 \end{bmatrix} \tag{8}$$

$$F_{\mathrm{out}}^j = \sum_j W_{\mathrm{lap}} * F_{\mathrm{in}}^j \tag{9}$$

where j is the index of channels for input and output features F_{in}^j and F_{out}^j, and $*$ is the convolution operation. We also apply multiple 3×3 convolutional layers to enlarge the receptive fields. Note that each convolution block is followed by a batch normalization (BN) layer and a nonlinear activation layer by default. Finally, we obtain the output feature $F_l^i \in \mathbb{R}^{P_i\times N\times N}$ for the ith CNN-based spatial encoder.

B. Feature Fusion-and-Separation Blocks

DSFE has encoded complementary features via the transformer and CNN. Now we introduce hierarchical F2SBs, each containing an HIGFM and an MSRM, to promote the mutual learning of modality-shared and modality-specific features in a progressive way. To this end, we introduce the deformable convolution in HIGFM which leverages the height information of LiDAR data to guide the fusion of dual-modality data. After that, we explore MSRM to separate and refine modality-specific features using the well fused ones. This fusion and separation procedure will repeat multiple times to realize deep feature fusion and enhancement.

Fig. 1 shows the pipeline of hierarchical F2SBs, where the skip connection is used to increase the detailed information of higher-level features. The whole computation process can be formulated as

$$\begin{aligned} X_s^2, X_h^2, X_l^2 &= f_{\mathrm{F2SB}_1}(X_s^1, F_h^1, F_l^1) \\ X_s^3, X_h^3, X_l^3 &= f_{\mathrm{F2SB}_2}(X_s^2, F_h^2, F_l^2) \\ X_s^4, X_h^4, X_l^4 &= f_{\mathrm{F2SB}_3}(\alpha\mathrm{Conv}(X_s^2) + \beta X_s^3, F_h^3, F_l^3) \end{aligned} \tag{10}$$

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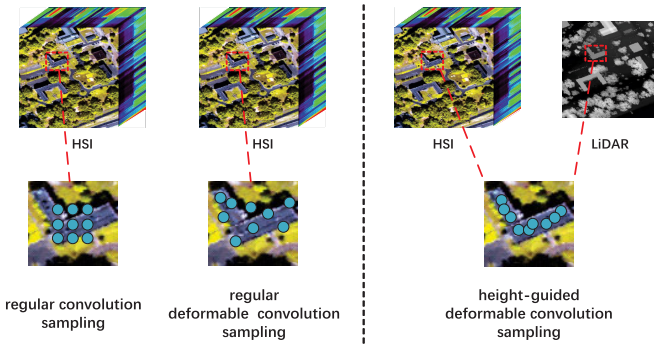


Fig. 2. Illustration of different convolution sampling ways used in prior methods (left) and ours (right).

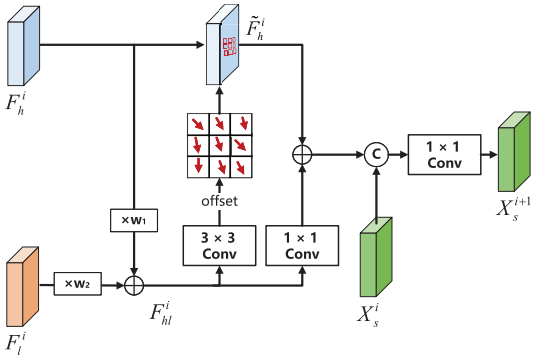


Fig. 3. Structure of HIGFM.

where f_{F2SB_i} denotes the operation of the ith F2SB ($i = 1, 2, 3$), $X_s^{i+1} \in \mathbb{R}^{P_i \times N \times N}$ is the modality-shared feature in each F2SB, and α and β are weight coefficients used for the skip connection. Note that, $X_s^1 \in \mathbb{R}^{P_1 \times N \times N}$ is an input tensor with all elements being zero, which is updated via convolution and concatenation operations and then propagated to the next F2SB for strengthening context information. α and β can be automatically learned and updated by the network (both are initially set to 0.5). The details of HIGFM and MSRM are introduced below.

1) Height Information Guided Fusion Module: Motivated by Dai et al. [47], we introduce the height-guided deformable convolution for better fusion of HSI and LiDAR data. The key idea of HIGFM is that we perform deformable convolution sampling on the HSI by using the height information contained in the LiDAR data. Fig. 2 compares different convolution sampling ways, including deformable convolution based ones [48], [49], and ours. The detailed structure of HIGFM is shown in Fig. 3, which is described as follows.

First, we perform the preliminary fusion of two complementary features in a dynamic fashion, which can be expressed as

$$F_{hl}^i = w_1 F_h^i + w_2 F_l^i \tag{11}$$

where w_1 and w_2 are learnable weights, and F_h^i and F_l^i are the output features of the ith transformer-based spectral encoder and CNN-based spatial encoder, respectively ($i = 1, 2, 3$).

Then, we learn the sampling location offsets from F_{hl}^i via 3×3 convolution and further extract the HSI feature map. For ease of understanding, the above operations for single-channel input feature maps are given as follows:

$$\tilde{F}_h^i(p_0) = \sum_{p_n \in R} w(p_n) F_h^i(p_0 + p_n + \triangle p_n) \tag{12}$$

$$\triangle p_n = \sum_{p_n' \in R} w'(p_n') F_{hl}^i(p_0 + p_n + p_n') \tag{13}$$

where $R = \{(-1, -1), (-1, 0), \ldots, (1, 1)\}$ is a regular 3×3 sampling grid, p_0 indicates a pixel position, w and w' are 3×3 convolution kernel weights, and $\triangle p_n$ is the offset of a sampling location. If $p_0 + p_n + \triangle p_n$ is fractional, its pixel value is obtained via bilinear interpolation.

Finally, we obtain the fused (or shared) feature through

$$X_s^{i+1} = \text{Conv}\big(\text{Concat}\big(\tilde{F}_h^i + \text{Conv}\big(F_{hl}^i\big), X_s^i\big)\big) \tag{14}$$

where X_s^i is the input shared feature in the ith F2SB, "Conv" denotes 1×1 convolution, and "Concat" denotes the feature concatenation along the channel dimension.

2) Modality Separation Refinement Module: This module aims to separate and refine modality-specific features from the fused ones. The resulting MSRM, together with HIGFM, forms a mutual-assistance learning mechanism for better feature representation. Fig. 4 shows the detailed structure of MSRM whose outputs can be formulated as

$$X_h^{i+1} = F_h^i + \text{CA}\big(\text{Conv}\big(X_s^{i+1} - F_h^i\big)\big) \tag{15}$$

$$X_l^{i+1} = F_h^i + \text{CA}\big(\text{Conv}\big(X_s^{i+1} - F_l^i\big)\big) \tag{16}$$

where "CA" denotes the channel attention module, F_h^i and F_l^i have been defined in (11), and X_s^{i+1} is the fused feature defined in (14) for $i = 1, 2, 3$. X_h^{i+1} (X_l^{i+1}) is the separate modality-specific feature, which will be subsequently fed into the $(i+1)$th spectral (spatial) encoder or the ED.

Specifically, we enhance the differential information by using the channel attention mechanism [50], [51]. As shown in Fig. 4, given the input $A_1^i \in \mathbb{R}^{P_i \times N \times N}$, we perform GAP to obtain the global feature $z_{\text{GAP}} \in \mathbb{R}^{P_i}$. Then, we feed z_{GAP} into a 1×1 convolutional layer, followed by a Sigmoid function,

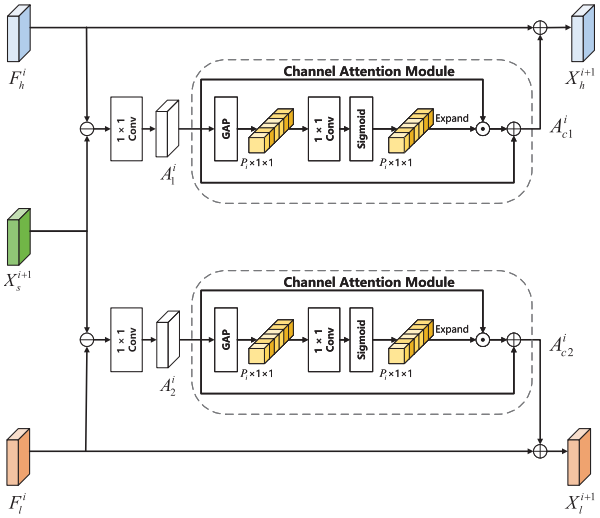


Fig. 4. Structure of MSRM.

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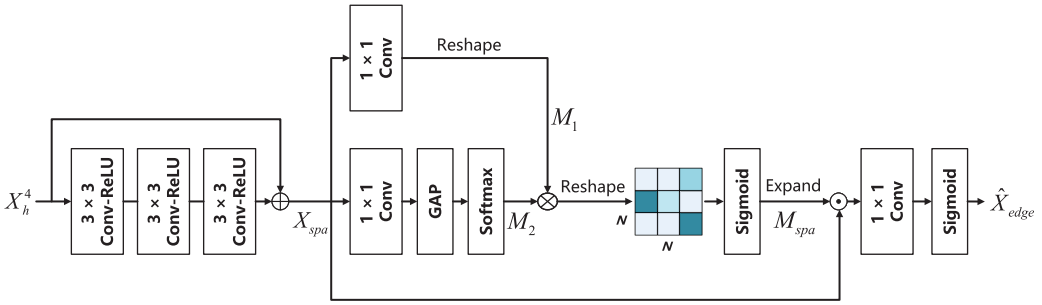


Fig. 5. Structure of ED used to predict the binary LiDAR edge map.

to compute the attention vector $z \in \mathbb{R}^{P_i}$, i.e.,

$$z = \text{Sigmoid}(\text{Conv}(z_{\text{GAP}})). \tag{17}$$

Next, we expand z via replication to make it and A_1^i have the same shape $P_i \times N \times N$. Finally, we perform elementwise multiplication and addition to obtain the channel attention enhanced feature $A_{c1}^i \in \mathbb{R}^{P_i \times N \times N}$, i.e.,

$$A_{c1}^i = \left(\text{Expand}(z) \odot A_1^i\right) + A_1^i \tag{18}$$

where $\odot$ denotes the elementwise multiplication. The feature of A_{c2}^i can be similarly obtained.

C. Edge Decoder

For the joint classification task, the HSI cube and LiDAR patch are usually fed into the network to classify the central pixel. This paradigm easily leads to misclassification of the central pixels in heterogeneous neighboring regions covering different land cover classes. To tackle this problem, we resort to an auxiliary edge prediction task and build a multitask learning (MTL) framework to improve the network's classification performance (see Fig. 1). As shown in [52] and [53], MTL can improve the model's generalization ability by performing multiple related tasks and exploring shared knowledge.

More specifically, we build an ED on the last output HSI feature to predict the LiDAR edge map (we consider that the edge information, which to some degree reflects the height information, can be easily and accurately extracted from the LiDAR image). Fig. 5 shows the structure of ED. Formally, given the input $X_h^4 \in \mathbb{R}^{S_4 \times N \times N}$, we first adopt a residual block, containing three 3×3 convolution layers and a residual-connection, to generate feature $X_{\text{spa}} \in \mathbb{R}^{S_4 \times N \times N}$

$$X_{\text{spa}} = f_{\text{res}}\left(X_h^4\right) \tag{19}$$

where f_{res} denotes the residual block. Then, we compute the spatial attention map $M_{\text{spa}} \in \mathbb{R}^{N \times N}$ as follows:

$$M_{\text{spa}} = \text{Sigmoid}\left(\text{Reshape}\left(M_2^T M_1\right)\right) \tag{20}$$

where T denotes matrix transposition, $M_1 \in \mathbb{R}^{S_4/2 \times (N \times N)}$ is the reshaped feature X_{spa} after 1×1 convolution, and $M_2 \in \mathbb{R}^{S_4/2}$ is a channelwise weight vector. Finally, we weight X_{spa} based on the spatial attention map, followed by a 1×1 convolution layer and a Sigmoid function to generate the predicted edge map $\hat{X}_{\text{edge}} \in \mathbb{R}^{N \times N}$ whose values range from 0 to 1

$$\hat{X}_{\text{edge}} = \text{Sigmoid}\left(\text{Conv}\left(\text{Expand}\left(M_{\text{spa}}\right) \odot X_{\text{spa}}\right)\right) \tag{21}$$

where $\odot$ denotes elementwise multiplication, and "Expand" performs the replication operation to make M_{spa} have S_4 channels. The output of ED, $\hat{X}_{\text{edge}}$, will be subsequently used to compute the edge prediction loss.

D. Overall Loss Function

We train our network by considering two tasks: the classification task and the edge prediction task. For the classification task, as shown in Fig. 1, we pass the modality-shared feature X_s^4 in the last F2SB to a 1×1 convolutional layer and then perform GAP along the channel dimension to obtain a global feature vector. Finally, we use a fully connected layer, followed by a Softmax function, for classification and the optimized object is the cross-entropy loss

$$\mathcal{L}_{\text{cls}} = -\frac{1}{M} \sum_{i=1}^{M} \boldsymbol{y}_i^T \log\left(\hat{\boldsymbol{y}}_i\right) \tag{22}$$

where M is the number of training samples, $\boldsymbol{y}_i$ is a one-hot label vector of the ith sample, and $\hat{\boldsymbol{y}}_i$ is the predicted label distribution.

For the edge prediction task, we apply the Canny edge detector on the input LiDAR patch X_l^1 to obtain binary edge map X_{edge} as the reference map. Then, we compute the edge prediction loss between the reference map X_{edge} and the predicted edge map $\hat{X}_{\text{edge}}$ with the Dice loss

$$\mathcal{L}_{\text{edge}} = 1 - \frac{2\sum_{i=1}^{N \times N} p_i \hat{p}_i}{\sum_{i=1}^{N \times N} p_i + \sum_{i=1}^{N \times N} \hat{p}_i}. \tag{23}$$

In (23), for a given pixel $x_i \in X_{\text{edge}}$, its pixel value $p_i = 1$ indicates that this pixel is an edge pixel, while $p_i = 0$ indicates that this pixel is not an edge pixel. For a given pixel $\hat{x}_i \in \hat{X}_{\text{edge}}$, its pixel value $\hat{p}_i$ is the prediction probability of this pixel being an edge pixel.

The whole end-to-end network is trained based on the following unified object function:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda \mathcal{L}_{\text{edge}} \tag{24}$$

where λ is a hyperparameter used to balance the two losses, which is set to 1 empirically. Note that, ED is only used in the training phase, and in the inference phase we only use the modality-shared stream to predict the classes of new samples.

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TABLE I
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET (BEST RESULTS ARE IN BOLD)

NO.	Class (Training/Testing)	CCR-Net	CoupledCNN	EndNet	FusAtNet	MDL-Cross	S^2ENet	MAHiDFNet	MFT	GLT-Net	Ours
1	Healthy grass (20/1231)	97.35	94.20	91.73	91.62	96.77	**99.56**	95.55	92.69	96.18	98.70
2	Stressed grass (20/1234)	85.14	**88.61**	83.50	84.46	85.02	87.16	86.30	87.20	86.27	86.86
3	Synthetis grass (20/677)	99.88	99.91	99.44	98.97	99.82	**99.97**	99.62	99.85	99.91	**99.97**
4	Tree (20/1224)	96.34	98.51	89.28	86.75	98.27	96.45	97.08	98.37	98.25	**98.82**
5	Soil (20/1222)	96.33	99.67	94.01	96.63	97.22	**99.98**	**99.98**	99.26	99.93	99.62
6	Water (20/305)	92.00	98.16	85.57	84.46	95.41	98.23	95.67	**100.00**	98.30	98.03
7	Residential (20/1248)	85.03	93.43	80.90	73.38	**98.45**	92.68	90.95	95.91	96.35	97.63
8	Commercial (20/1224)	84.79	90.77	74.97	74.87	86.32	85.87	81.76	85.78	90.46	**95.18**
9	Road (20/1232)	68.56	82.76	69.20	70.83	83.91	78.65	68.93	85.15	90.94	**91.02**
10	Highway (20/1207)	82.57	88.62	54.83	78.36	80.56	85.20	73.01	**93.12**	90.74	92.26
11	Railway (20/1215)	75.92	96.81	78.98	82.19	82.45	86.73	72.61	90.37	96.97	**97.35**
12	Park lot 1 (20/1213)	86.60	88.95	61.78	85.29	83.58	92.83	88.66	91.26	**93.22**	92.83
13	Park lot 2 (20/449)	76.66	97.37	50.91	77.37	96.70	94.79	94.03	96.66	**99.24**	98.22
14	Tennis court (20/408)	91.03	**100.00**	96.91	99.66	95.69	99.90	**100.00**	**100.00**	**100.00**	**100.00**
15	Running track (20/640)	97.56	99.62	99.06	97.28	99.66	**100.00**	99.66	**100.00**	**100.00**	**100.00**
	OA	87.01	93.42	79.73	84.19	90.76	91.91	87.66	93.16	94.89	**95.77**
	AA	87.72	94.52	80.74	85.47	91.99	93.20	89.61	94.38	95.78	**96.43**
	Kappa	85.95	92.88	78.10	82.92	90.01	91.26	86.66	92.61	94.48	**95.42**

TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE TRENTO DATASET (BEST RESULTS ARE IN BOLD)

NO.	Class (Training/Testing)	CCR-Net	CoupledCNN	EndNet	FusAtNet	MDL-Cross	S^2ENet	MAHiDFNet	MFT	GLT-Net	Ours
1	Apple trees (20/4014)	98.44	98.99	75.11	97.84	93.80	98.84	**99.24**	97.77	99.21	98.69
2	Buildings (20/2883)	98.43	96.69	98.03	97.37	93.53	97.23	93.84	98.39	**98.93**	98.33
3	Ground (20/459)	94.34	99.56	95.51	99.83	92.07	99.52	98.74	99.13	96.21	**100.00**
4	Woods (20/9103)	99.68	99.09	97.98	99.61	99.94	99.01	94.22	**100.00**	**100.00**	**100.00**
5	Vineyard (20/10481)	97.89	99.10	72.69	99.17	99.38	99.37	93.95	99.83	**99.99**	99.76
6	Roads (20/3154)	86.39	89.63	85.64	80.56	87.58	82.17	83.18	93.28	93.61	**97.61**
	OA	97.30	97.87	84.79	97.01	96.89	97.18	93.67	98.77	99.06	**99.34**
	AA	95.86	97.18	87.49	95.73	94.38	96.02	93.86	98.07	97.99	**99.09**
	Kappa	96.40	97.15	80.13	96.01	95.84	96.24	91.60	98.36	98.75	**99.12**

III. EXPERIMENTS AND PERFORMANCE EVALUATION

A. Datasets and Experimental Setting

To verify the effectiveness of the proposed network, we conduct experiments on three widely used HSI and the associated LiDAR datasets. The details are introduced below.

1) Houston2013 Dataset: The Houston2013 Dataset was acquired by the Compact Airborne Spectrographic Imager (CASI) sensor, covering the University of Houston campus and adjacent areas. This dataset was published by the 2013 Geoscience and Remote Sensing Society (GRSS) Data Fusion Competition [54]. The HSI and LiDAR data have the same spatial size of 349 × 1905 pixels and the same spatial resolution of 2.5 m. There are 144 bands in the HSI with wavelengths ranging from 0.38 to 1.05 μm. This dataset, including a total of 15 029 ground-truth samples and 15 classes of land covers, is available online from the IEEE GRSS Data website: http://dase.grss-ieee.org/. The number of training and testing samples in each class is shown in Table I.

2) Trento Dataset: The Trento Dataset was collected by the AISA Eagle sensor in southern Trento, Italy. This dataset contains HSI and LiDAR data, and both have a spatial size of 166 × 600 pixels and a spatial resolution of 1 m. The HSI has 63 spectral bands with wavelengths ranging from 0.42 to 0.99 μm. There are a total of 30 214 ground-truth samples and six classes including buildings, ground, apple trees, vineyard, woods and roads. The number of training and testing samples in each class is shown in Table II.

3) MUUFL Dataset: The MUUFL dataset [55], [56] was collected in November 2010 at the University of Southern Mississippi's Gulf Park campus in Long Beach, Mississippi, USA. The HSI has a spatial size of 325 × 220 pixels with 64 available spectral bands ranging from 0.38 to 1.05 μm, and the spatial resolution is 0.54 × 1.0 m. The LiDAR image has the same spatial size while its spatial resolution is 0.60 × 0.78 m. There are 53 687 ground-truth samples and 11 classes of land covers. The number of training and testing samples in each class is shown in Table III.

We conduct experiments with the PyTorch framework on a workstation with an Intel Silver 4210R 2.40-GHz processor and two NVIDIA GEFORCE RTX 3080 10-GB GPUs. We use Adam [57] as the optimizer by setting the optimum learning rate as 0.0008. The batch size of input data during training is 64 and the number of training epochs is 300. For Houston2013 and MUUFL datasets, we set $S_1 = 36$, $S_2 = 64$, $S_3 = 96$, $P_1 = 64$, $P_2 = 96$, and $P_3 = 96$. For Trento dataset, we set

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TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE MUUFL DATASET (BEST RESULTS ARE IN BOLD)

NO.	Class (Training/Testing)	CCR-Net	CoupledCNN	EndNet	FusAtNet	MDL-Cross	S^2ENet	MAHiDFNet	MFT	GLT-Net	Ours
1	Trees (20/23226)	83.12	**88.96**	78.75	77.83	84.02	87.79	88.56	85.92	87.73	88.57
2	Mostly grass (20/4250)	67.24	73.68	69.28	69.60	56.41	81.56	67.56	67.93	**81.67**	78.86
3	Mixed ground surface (20/6862)	59.33	58.10	54.90	68.60	64.02	55.71	64.25	70.10	65.84	**71.16**
4	Dirt and sand (20/1806)	78.08	86.20	64.55	79.28	84.51	93.96	83.22	**94.91**	90.59	85.19
5	Road (20/6667)	79.41	80.20	86.64	77.40	60.55	83.35	**87.69**	72.66	86.89	84.61
6	Water (20/446)	98.70	99.09	98.43	86.50	94.89	99.10	44.87	99.33	99.51	**99.55**
7	Building shadow (20/2213)	82.41	81.77	86.80	85.43	**93.46**	90.60	91.90	90.06	90.78	89.13
8	Building (20/6220)	93.03	94.64	87.45	82.62	95.18	89.68	81.85	92.14	94.85	**96.23**
9	Sidewalk (20/1365)	46.97	62.67	58.43	31.56	57.77	41.98	**74.72**	54.21	59.53	63.54
10	Yellow curb (20/163)	64.29	81.52	84.05	57.42	83.07	71.66	80.73	71.17	76.44	**85.28**
11	Cloth panels (20/249)	95.82	94.77	88.67	87.63	97.83	97.35	97.52	**99.60**	97.19	96.84
	OA	78.51	81.86	76.50	75.73	77.52	82.09	78.44	81.32	84.77	**85.29**
	AA	77.13	80.23	78.00	73.08	79.25	81.16	76.52	81.64	84.24	**85.15**
	Kappa	72.58	76.21	70.29	69.32	71.53	76.99	71.25	76.19	80.44	**81.07**

$S_1 = 16$, $S_2 = 32$, $S_3 = 48$, $P_1 = 32$, $P_2 = 48$, and $P_3 = 48$. The reason for using these different parameter settings is that the Trento dataset contains fewer classes of land covers than the other two datasets and a small number of learnable parameters are enough to achieve good classification results. Note that we always set $S_1 = 1$ for the input LiDAR patch. We adopt the Kappa coefficient, overall accuracy (OA), and average accuracy (AA) to evaluate the classification performance.

B. Comparison With State-of-the-Art Methods

We compare the state-of-the-art methods including cross-channel reconstruction network (CCR-Net) [22], coupled convolutional neural network (CoupledCNN) [58], encoder-decoder network (EndNet) [59], FusAtNet [27], multimodal deep learning-cross fusion (MDL-Cross) [60], S^2ENet [26], MAHiDFNet [41], MFT [37], and global-local transformer network (GLT-Net) [53]. Among them, EndNet is based on encoder–decoder networks, while CCR-Net, CoupledCNN and MDL-Cross are based on CNNs. FusAtNet, S^2ENet and MAHiDFNet are attention-based CNN methods, while MFT and GLT-Net are transformer-based methods. We implement the compared methods based on the published codes provided by the authors with default parameter settings. Specifically, the learning rates for CoupledCNN, EndNet, MDL-Cross, S^2ENet, CCR-Net, FusAtNet, MAHiDFNet, MFT and GLT-Net are 0.001, 0.001, 0.001, 0.001, 0.001, 0.000005, 0.0001, 0.00005 and 0.0005, respectively; the batch sizes are 64, 64, 64, 64, 32, 64, 100, 64 and 64, respectively; the numbers of training epochs are 200, 250, 250, 128, 200, 1000, 200, 500 and 500, respectively. We run five independent experiments and take the average values as the final classification results.

Tables I–III present the quantitative results in terms of the classification accuracy per class, OA, AA, and the Kappa coefficient. The best result in each specific row is shown in bold type. First, EndNet which employs the fully connected layers performs poorly on all the three datasets. CCR-Net and MDL-Cross, both of which employ CNNs, generally show better classification results than EndNet. Second, for attention-based CNN methods, S^2ENet outperforms FusAtNet and MAHiDFNet. Also, S^2ENet shows much better results than CCR-Net and MDL-Cross on the Houston2013 and MUUFL datasets, demonstrating the efficacy of cross-modal attention mechanism. Third, CoupledCNN, a CNN model based on decision-level fusion, achieves better results than S^2ENet on the Houston2013 and Trento datasets, and shows comparable results on the MUUFL dataset. Fourthly, for transformer-based models, GLT-Net outperforms MFT by utilizing the multiscale information fusion and image reconstruction task. By contrast, our proposed method explores hierarchical fusion-and-separation blocks to promote mutual learning between modality-shared and modality-specific features. Meanwhile, our proposed method builds an edge prediction task to strengthen dual-modality information interaction. As can be seen in Tables I–III, our method yields the best classification performance on all the three datasets. Finally, to determine whether the performance improvement of the proposed method over GLT-Net is statistically significant or not on the Trento and MUUFL datasets, we perform the paired Student's t-tests. As shown in Fig. 6, our method leads to significant improvement over GLT-Net at the significance level $\alpha = 0.05$.

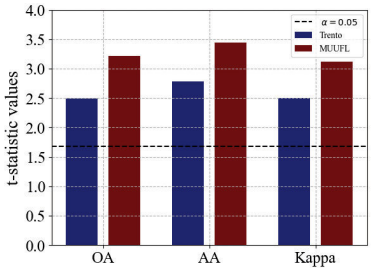


Fig. 6. t-statistic values of paired t-tests between GLT-Net and the proposed model on the Trento and MUUFL datasets, which are obtained by using the same 50 random splits of the training and testing sets for each dataset. The dashed line indicates the corresponding threshold $t_{0.05}(49) = 1.677$ at the significance level $\alpha = 0.05$.

Figs. 7–9 show the classification maps on the three datasets. One can see that, EndNet wrongly classifies many pixels relying on the fully connected layers and produces a noise-like classification map. S^2ENet reduces noise-like results by exploring the attention mechanism. MFT and GLT-Net give relatively poor classification results for the

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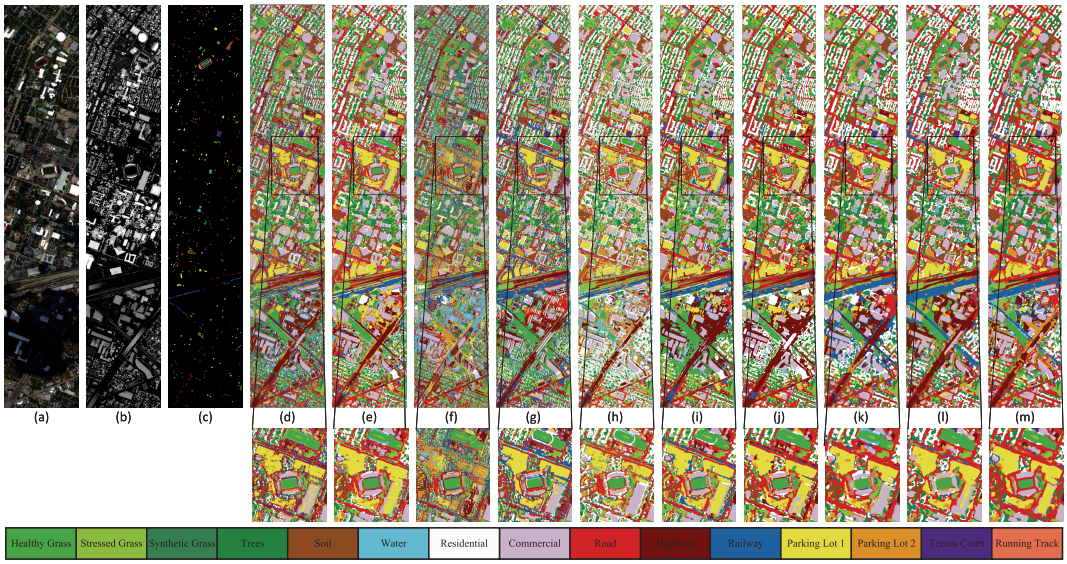


Fig. 7. Classification maps of the Houston2013 dataset. (a) Pseudocolor image for HSI. (b) LiDAR image. (c) Ground-truth map. (d) CCR-Net. (e) CoupledCNN. (f) EndNet. (g) FusAtNet. (h) MDL-Cross. (i) S^2ENet. (j) MAHiDFNet. (k) MFT. (l) GLT-Net. (m) Ours.

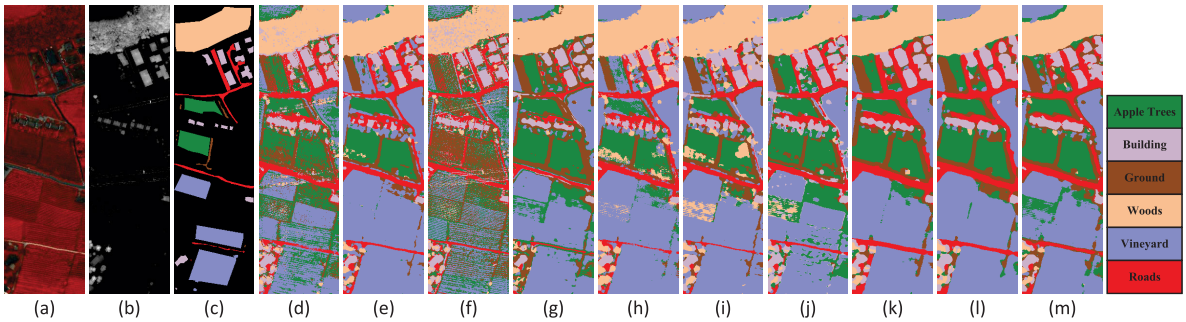


Fig. 8. Classification maps of the Trento dataset. (a) Pseudocolor image for HSI. (b) LiDAR image. (c) Ground-truth map. (d) CCR-Net. (e) CoupledCNN. (f) EndNet. (g) FusAtNet. (h) MDL-Cross. (i) S^2ENet. (j) MAHiDFNet. (k) MFT. (l) GLT-Net. (m) Ours.

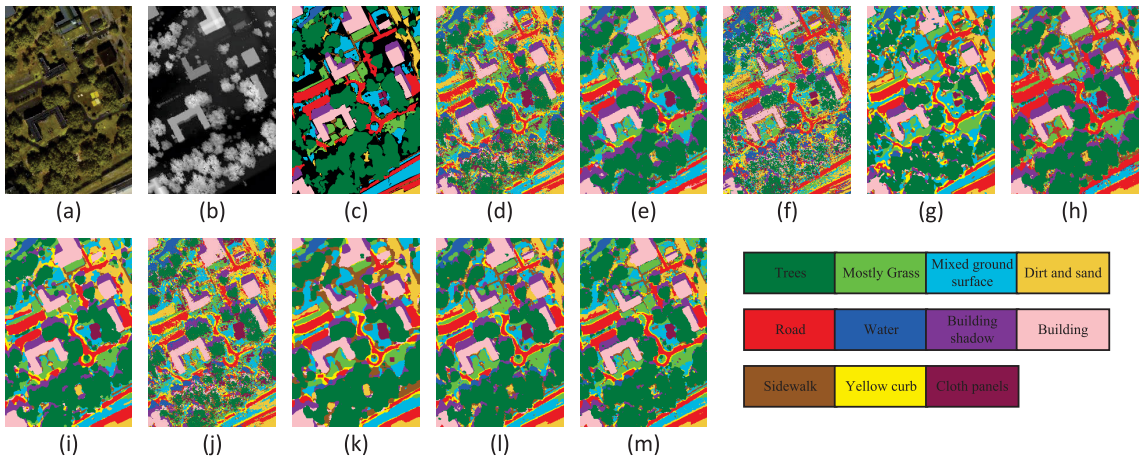


Fig. 9. Classification maps of the MUUFL dataset. (a) Pseudocolor image for HSI. (b) LiDAR image. (c) Ground-truth map. (d) CCR-Net. (e) CoupledCNN. (f) EndNet. (g) FusAtNet. (h) MDL-Cross. (i) S^2ENet. (j) MAHiDFNet. (k) MFT. (l) GLT-Net. (m) Ours.

classes of "commercial" on the Houston2013 dataset and "Yellow curb" on the MUUFL dataset. By comparison, our proposed method produces better classification maps, which are also confirmed by the quantitative results shown in Tables I–III.

C. Analysis of Our Model

1) Impact of Spatial Size: As shown in Fig. 10, when we set the spatial size as 9×9, 9×9, and 7×7 for the Houston2013, the Trento, and the MUUFL dataset,

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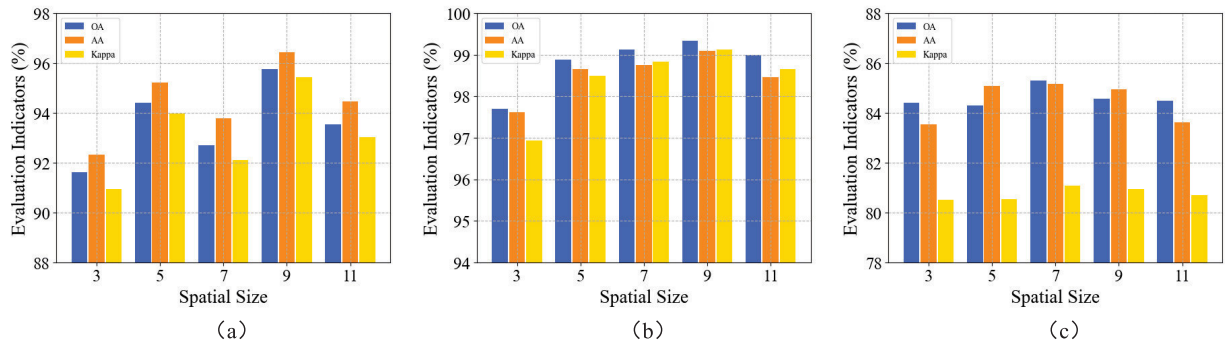


Fig. 10. Impact of the spatial size on the classification performance. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

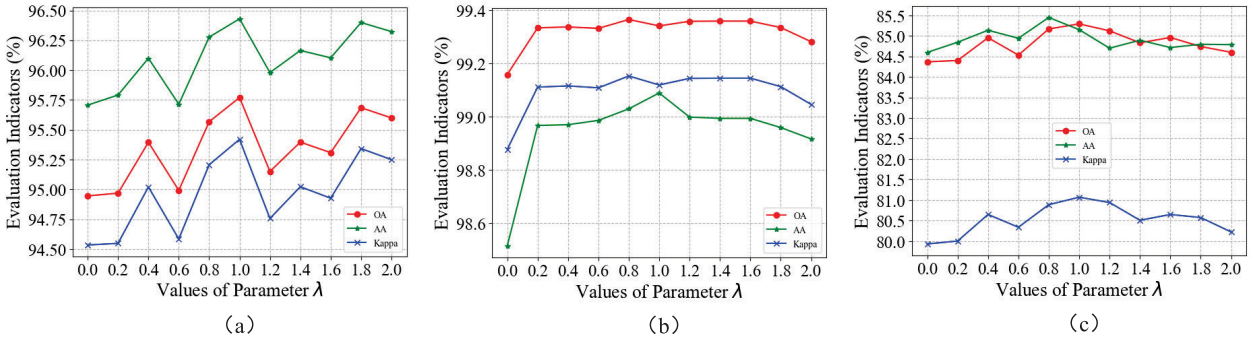


Fig. 11. Impact of the parameter λ on the classification performance. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

TABLE IV
IMPACT OF PARALLEL GCS ON THE CLASSIFICATION PERFORMANCE

#Parallel GCs	#Groups in each GC	Houston2013			Trento			MUUFL		
		OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
1	1	94.64	95.56	94.21	99.29	98.73	99.05	83.49	85.05	78.88
	2	95.25	95.96	94.86	98.67	98.47	98.23	83.43	82.27	78.62
	4	95.70	96.25	95.35	99.33	99.00	99.10	83.47	84.34	80.31
2	1, 2	94.83	95.67	94.41	99.32	98.75	99.09	83.49	83.33	78.82
	1, 4	94.88	95.75	94.46	99.28	98.77	99.04	84.98	84.30	80.67
	2, 4	94.98	95.85	94.58	98.75	98.59	98.34	85.11	83.83	80.93
3	1, 1, 1	93.75	93.76	93.16	98.54	98.34	98.06	84.89	84.98	80.65
	2, 2, 2	94.45	95.47	93.92	99.11	98.78	98.82	84.59	84.04	80.19
	4, 4, 4	94.67	95.33	94.08	99.02	98.60	98.69	84.82	83.38	80.56
	1, 2, 4	**95.77**	**96.43**	**95.42**	**99.34**	**99.09**	**99.12**	**85.29**	**85.15**	**81.07**
4	1, 2, 4, 1	94.26	95.08	93.79	98.96	98.31	98.61	84.41	81.53	79.62
	1, 2, 4, 2	93.48	94.36	92.95	99.30	98.90	99.06	84.50	84.24	79.38
	1, 2, 4, 4	94.79	95.24	94.37	98.80	98.24	98.40	83.71	84.41	79.07

respectively, our model achieves the best classification performance in terms of OA, AA and Kappa coefficient. As the Houston2013 and Trento datasets have a lower spatial resolution than the MUUFL dataset, these two datasets need larger image cubes/patches to train the deep learning network for better classification performance. By default, we choose the spatial size of 9×9 for the Houston2013 and Trento datasets, and 7×7 for the MUUFL dataset.

2) Impact of Parameter λ: The parameter λ in (24) tunes the weight of the edge prediction loss. As shown in Fig. 11, different datasets have varying sensitivity to λ. When λ is varied from 0 to 2, the three evaluation indicators (i.e., OA, AA, Kappa) can fluctuate by about 0.9%, 0.6%, and 1% on the Houston2013, the Trento, and the MUUFL dataset, respectively. More classes of land covers with complex distributions in the Houston2013 and MUUFL datasets account for the above relatively large fluctuations. In particular, when we set λ as 0, the classification performance of our model can be obviously reduced on the three datasets. This demonstrates that the edge prediction loss helps to learn good feature representations and improve the classification performance. In our experiments, we have chosen λ as 1 by default which can generally obtain good performance on all the three datasets.

3) Impact of Parallel GCs: Table IV shows the impact of parallel GCs on the classification performance (note that, since the number of spectral channels for the Trento dataset is 63, we add an all-zero channel to make it suitable for

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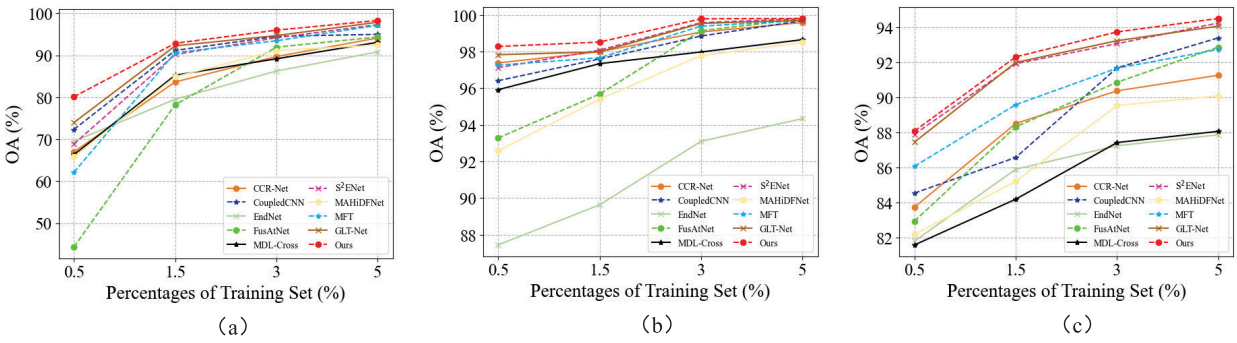


Fig. 12. Classification performance obtained by varying the percentage of training set. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

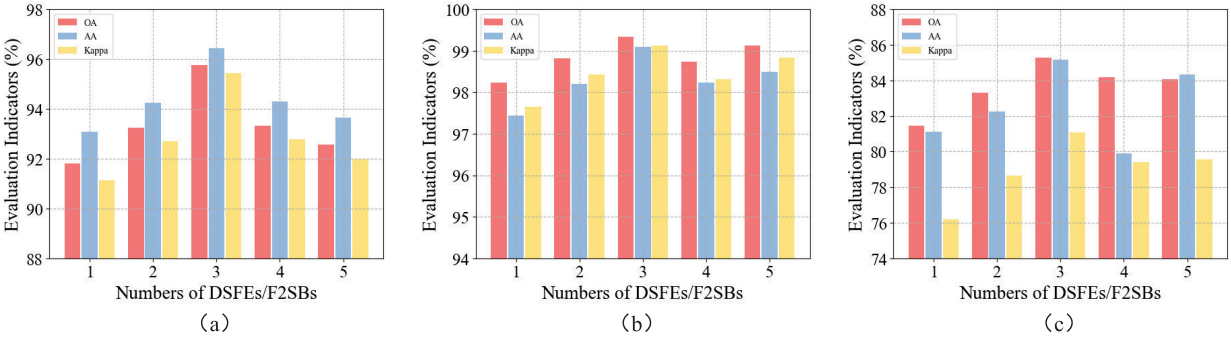


Fig. 13. Impact of different numbers of DSFEs/F2SBs on the classification performance. (a) Houston2013 dataset. (b) Trento dataset. (c) MUUFL dataset.

TABLE V
IMPACT OF THE VALUE OF L ON THE CLASSIFICATION PERFORMANCE

L	Houston2013			Trento			MUUFL		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
1, 1, 1	93.78	94.65	93.28	98.06	98.05	98.04	85.17	85.05	78.10
2, 2, 2	94.96	95.63	94.55	98.32	98.97	98.10	84.00	84.02	79.43
2, 2, 4	93.82	94.71	93.32	99.03	98.19	98.24	84.70	82.51	80.25
2, 2, 6	**95.77**	**96.43**	**95.42**	**99.34**	**99.09**	**99.12**	**85.29**	**85.15**	**81.07**
2, 2, 8	93.51	94.45	92.98	98.82	98.11	98.43	84.90	81.23	80.69
3, 3, 3	93.45	94.43	92.92	99.18	98.79	98.91	84.87	84.68	80.55
4, 4, 4	94.50	95.20	94.05	98.99	98.36	98.66	85.02	81.25	80.62

TABLE VI
IMPACT OF FILTERS USED IN (8) ON THE CLASSIFICATION RESULTS

Filter	Houston2013			Trento			MUUFL		
	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Sobel-1	93.90	94.78	93.42	99.07	98.55	98.75	84.32	84.23	79.97
Sobel-2	93.87	94.62	93.57	99.18	98.80	98.90	84.16	84.69	79.74
Laplacian	**95.77**	**96.43**	**95.42**	**99.34**	**99.09**	**99.12**	**85.29**	**85.15**	**81.07**
Gaussian	94.06	94.99	93.58	99.30	98.97	99.07	84.27	84.87	79.83

grouping). As can been seen, the use of more parallel GCs does not mean higher classification accuracies. For example, the performance obtained by using four parallel GCs can be worse than that obtained by using three parallel GCs. Overall, the use of three parallel GCs with diverse group settings for different GCs [i.e., (1, 2, 4)] achieves the best classification results.

4) Impact of Different Percentages of Training Set: We randomly select different percentages (ranging from 0.5% to 5%) of samples from each class for training, and the rest are used for testing. As shown in Fig. 12, when the percentage of training set is 0.5%, our method still surpasses other ones in terms of OA on all the three datasets, indicating its excellent classification performance with limited training samples.

5) Impact of Different Numbers of DSFEs/F2SBs and L: We evaluate the impact of different numbers of DSFEs/F2SBs and repetitions of encoding in the transformer-based spectral encoder (i.e., the value of L) on the classification performance. As shown in Fig. 13, when the number of DSFEs/F2SBs increases from 1 to 3, the classification performance is steadily improved. However, when the number of DSFEs/F2SBs is larger than 3, the classification performance is reduced due to the overfitted model. Hence, the optimal number is 3. Table V further shows the impact of the value of L. As can be seen, too large or too small values of L lead to suboptimal results. By default, we set L as 2, 2, 6 for three consecutive transformer-based spectral encoders to obtain the best results.

6) Impact of the Laplace Filter Used in (8): For this experiment, we replace the Laplace filter (LF) with Sobel and Gaussian filters. Table VI shows the classification results, where Sobel-1 means the Sobel horizontal edge filter and Sobel-2 means the gradient magnitude of Sobel filters. One can see that the LF obtains the best classification results. The reasons are as follows. Sobel-1 is a directional derivative filter, which is sensitive to different orientations of edges. The use of Sobel-1 can wrongly classify land covers in varying backgrounds. Sobel-2 preserves the difference-magnitude information but loses the important

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TABLE VII
IMPACT OF DIFFERENT COMPONENTS USED IN OUR METHOD

	Components				Houston2013			Trento			MUUFL		
	LF	HIGFM	MSRM	ED	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
Baseline					89.67	91.33	88.83	98.77	98.17	98.36	80.03	81.28	74.39
+ LF	√				93.16	94.30	92.61	99.01	98.16	98.68	82.30	82.18	77.46
+ LF + MSRM	√		√		94.02	95.04	93.54	98.96	98.44	98.07	82.53	83.66	77.67
+ LF + HIGFM	√	√			94.72	95.63	94.30	99.05	98.49	99.02	82.94	82.47	78.23
+ LF + HIGFM + MSRM	√	√	√		94.95	95.71	94.53	99.16	98.52	98.88	84.36	84.59	79.93
+ LF + HIGFM + MSRM + ED†	√	√	√	√	94.94	95.62	94.52	99.29	99.07	98.82	85.02	85.08	80.74
+ LF + HIGFM + MSRM + ED	√	√	√	√	**95.77**	**96.43**	**95.42**	**99.34**	**99.09**	**99.12**	**85.29**	**85.15**	**81.07**

TABLE VIII
COMPARISON OF THE NUMBER OF PARAMETERS, FLOPS, AND TRAINING/TESTING TIME

Dataset	Complexity	CCR-Net	CoupledCNN	EndNet	FusAtNet	MDL-Cross	S^2ENet	MAHiDFNet	MFT	GLT-Net	Ours
Houston2013	#Parameters	71.23K	104.78K	91.97K	36.90M	119.10K	270.80K	77.56M	312.43K	558.72K	1.19M
	#FLOPs (M)	0.35	2.48	0.46	7273.79	6.72	65.43	156.38	50.32	312.28	195.33
	Training time (s)	1.08	1.29	0.47	2.66	1.12	1.48	3.90	1.82	1.56	1.47
	Testing time (s)	1.53	2.52	1.68	5.05	2.83	1.26	16.82	3.99	8.62	4.24
Trento	#Parameters	59.27K	101.30K	88.71K	36.24M	106.85K	177.19K	74.16M	262.89K	394.30K	0.52M
	#FLOPs (M)	0.29	1.24	0.44	7115.70	5.57	42.85	120.52	25.30	186.72	84.70
	Training time (s)	0.38	0.32	0.33	0.82	0.43	0.56	1.14	0.66	0.51	0.44
	Testing time (s)	2.14	3.26	2.44	11.26	3.48	2.87	19.23	5.26	15.28	6.45
MUUFL	#Parameters	59.58K	103.23K	89.07K	36.25M	107.32K	178.50K	74.13M	248.10K	396.81K	1.18M
	#FLOPs (M)	0.30	1.21	0.44	7117.66	5.59	43.13	110.47	21.82	188.27	117.62
	Training time (s)	0.62	0.59	0.37	1.84	0.65	1.02	2.12	1.11	0.95	0.72
	Testing time (s)	3.56	4.15	3.84	17.83	4.59	4.84	25.68	9.05	20.13	13.64

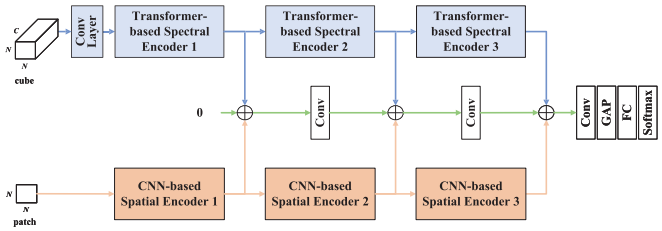


Fig. 14. Pipeline of the baseline method in our ablation study.

difference-sign information. The Gaussian filter can smooth image details, without highlighting edge information. By contrast, the adopted LF is isotropic and contains important difference-sign information, which helps to extract discriminative edge features and encode the height information of LiDAR images.

7) Impact of Different Components: For this experiment, we define the following baseline method: we replace the LF in our CNN-based spatial encoder with an ordinary convolution, replace F2SB with the elementwise addition operation, and remove the ED branch (see Fig. 14). As can be seen from Table VII, each component contributes to the classification performance. By combining all the components, our method achieves the best results on the three datasets. In addition, we replace the Dice loss used in (23) with the binary cross-entropy loss, and denote the resulting module as ED†. We see that the use of Dice loss obtains better classification results. This is because the Dice loss can alleviate the imbalance issue between edge and nonedge classes.

8) Complexity Analysis: We analyze the complexity of different methods by comparing the number of model parameters, the number of floating-point operations (FLOPs), and training/testing time. As can be seen from Table VIII, transformer-based MFT and GLT-Net have more model parameters than FC-based EndNet, and CNN-based CCR-Net, CoupledCNN, MDL-Cross, and S^2ENet. The attention-based MAHiDFNet and FusAtNet have the most model parameters (more than 74M and 36M, respectively) on the three datasets, followed by the proposed model (about 1M on Houston2013 and MUUFL, and 0.5M on Trento). In addition, FusAtNet has the most FLOPs, followed by GLT-Net and our model. Table VIII also reports the average training time for each epoch and the testing time for all testing samples. MAHiDFNet, FusAtNet and GLT-Net require more training time than others, followed by S^2ENet, MFT and our method. For the testing time, our method takes less time than MAHiDFNet, GLT-Net and FusAtNet. Therefore, the proposed method achieves the best classification performance with moderate FLOPs, model parameters and training/testing time among the compared methods in the experiment.

9) Feature Visualization: Fig. 15 visualizes the original HSI/LiDAR features and the output features of the last fully connected layer in GLT-Net and HFSNet via t-distributed stochastic neighbor embedding (t-SNE) [61]. One can see that the original HSI and LiDAR features are confusing, especially on the Houston2013 and MUUFL datasets. Compared with GLT-Net, our HFSNet shows better separability among distinct classes, even though our model is trained with only 20 samples per class. Hence, our model generalizes well to learn

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Fig. 15. Feature visualization via t-SNE for testing samples in Houston2013, Trento and MUUFL datasets (from left to right). (a)–(c) Original HSI features. (d)–(f) Original LiDAR features. (g)–(i) GLT-Net features. (j)–(l) HFSNet features. Different land-cover classes are plotted in different colors.

discriminative features for jointly classifying HSI and LiDAR data with limited training samples.

IV. Conclusion

In this article, we have proposed a height information guided hierarchical fusion-and-separation network called HFSNet for joint classification of HSI and LiDAR data. In HFSNet, we design DSFEs to excavate the complementary spectral sequence information from HSIs and spatial information from LiDAR data. In spatial encoders, we embed the Laplacian filter to explicitly encode the height (elevation) information contained in the LiDAR data. Furthermore, we propose a deformable convolution-based HIGFM module and an MSRM module to promote the mutual learning of modality-shared and modality-specific features in a deeply interactive way. Additionally, we incorporate an auxiliary ED into our model to predict the LiDAR edge map from the HSI feature to perform task-level cross-modality information interactions and improve the model's generalization ability. Experimental results on three datasets show the superiority of HFSNet to the state-of-the-art methods for joint classification of HSI and LiDAR data with limited training samples.

HFSNet still has some limitations that remain to be addressed in the future work. For example, some samples belonging to different classes are still inseparable according to the visualization results. Moreover, the joint classification model of HFSNet requires paired HSI and LiDAR data for training and testing. How to train a classification model with unpaired multimodal data and successfully classify samples with missing modalities is worth further exploring.

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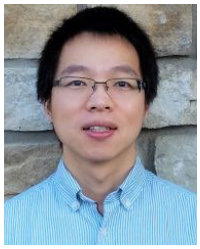
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